

1

Proposed Methodology

Deepfake Audio Detection

ASVspoof 2019 LA | Bona fide / Spoof | Known + unseen attacks

2

Preprocessing

Mono conversion | 16 kHz resampling | Normalization | Padding + masking

3

Augmentation

RawBoost ① + ② | Codec augmentation | Real-world distortions

4

Teacher Model

Wav2Vec2-XLSR | AASIST | P2SGrad / OC-Softmax

5

Soft Predictions

Teacher confidence scores | Soft labels

6

Student Model

Frozen Wav2Vec2-XLSR | AASIST-L

7

Knowledge Distillation

Hard label loss | KD loss | Knowledge transfer

8

Prediction

Bona fide | Spoof

9

Evaluation

EER, min-tDCF | Accuracy, Precision | Recall, F1-score | Confusion matrix